Matteo Paolini

Mechanical engineer · mechanical design, FEA, manufacturing & robotics

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Available from February 2027

EXPERIENCE

Research Intern — robotic manipulation for demining · Royal Military Academy, Brussels — 2026 – present

- Humanitarian demining: simulating cluttered scenes in MuJoCo and developing a learning-based policy that decides which objects to remove, and how, to extract a PFM-1 landmine safely; next: UR5 arm and transfer to the real robot.

Machinery Maintenance Assistant — seasonal · PM Macchine, Corropoli (IT) — Summers 2016 – 2020

- Maintenance of leather-goods machinery (Adler, ATOM, Galli, OMAC) with experienced technicians or independently: sewing-machine hook timing, pneumatic troubleshooting, hydraulic seals, faulty sensors and control boards. Occasional support ongoing.
- Designed and 3D-printed many custom fixtures; most recent: a fixture positioning trouser labels during sewing, in PLA with a high wall count for stiffness.

Assembly & Machine Calibration Intern · Co.Me.Ta. Srl, Montegranaro (IT) — 2019

Assembled machine frames and integrated sensors, motors and electromechanical parts into industrial machines; final functional calibration.

Assembly & Wiring Intern · Galli S.p.A., Vigevano (IT) — assembled and wired leather-goods machinery — 2018

EDUCATION

M.Sc. Mechanical Engineering — Mechatronics · UNIVPM, Ancona · Erasmus at ULB & VUB, Brussels (2025–26) — 2024 – Feb 2027

Composite structures, manufacturing, virtual prototyping, robotics, control system design, industrial automation.

B.Sc. Mechanical Engineering · UNIVPM, Ancona — 2021 – 2024

Thesis: collision-free pick-and-place planning for the redundant 7-DOF TIAGo arm (MATLAB).

SELECTED PROJECTS

Soft tentacle gripper · VUB Brussels · team of 5, supervised by Prof. B. Vanderborght — 2025

- Owned the tentacle: compared three chamber layouts in ANSYS with a 2nd-order **Ogden hyperelastic** model of Dragon Skin 20, rejecting one that bulged instead of bending; used the FEA to rank designs given a node-limited student license.
- Selected polymer casting from a process screening; designed the two-part 3D-printed mold with removable core. Lesson learned: core misalignment varied wall thickness — next step tighter-tolerance molds and pressure casting.
- Modular gripper with individually replaceable tentacles on a 2-DOF arm (belt reductions from a 2.40 N·m torque requirement); grasped and lifted an apple.

Apple Watch case — production study · UNIVPM · team of 2 — 2025

- Reconstructed the Al 7005-T6 case in Siemens NX with functional **GD&T** (±0.01 mm fits, Ø0.05 concentricity on the Ø3 H7 crown bore).
- Process route from laser blanking (608 blanks per panel) through cavity and contour milling in a custom vise to vibratory finishing (Ra 0.8 → 0.4 µm).
- Sized, on paper, a line for **1,000,000 parts/year** (1 laser, 89 CNC centers, 1 tumbler) from MRR and cutting power; removed a milling bottleneck by scheduling the fleet over four periods; Taylor tool-life and cost-per-part model.

vacUUUm — cordless vacuum, concept to drawings · Virtual Prototyping, UNIVPM · team project — 2025

- Turned 13 user needs into 11 measurable requirements and a modular architecture of 7 subsystems; modeled every subsystem in **Siemens NX** with exploded assemblies and production drawings.
- Checked right- and left-handed postures with digital human models in Tecnomatix (RULA) before freezing the design.

Dual-arm object handover · Robotics II, VUB · academic project — 2025 – 2026

- Two Franka Panda arms pass a cube mid-air in Drake: handover pose from the arms' workspace intersection, IK, collision-aware planning and a 13-state coordinator.

Also: 1U CubeSat CFRP/aluminum structure (ANSYS ACP, in progress) · LPBF 316L metallography and Vickers testing · PLC elevator controller (GRAFCET) · bearing-fault diagnosis from motor current · fault diagnosis on a multiphase plant.

SKILLS

Mechanical	Siemens NX · SolidWorks · ANSYS Workbench / ACP · hyperelastic & composite FEA · GD&T & drawings · DFM · mold design · process planning
Manufacturing	CNC process planning · cutting parameters & tool life · cost modeling · silicone casting · FDM 3D printing · metallography
Robotics & code	Python · MuJoCo · Drake · MATLAB / Simulink · Arduino · Tecnomatix
Languages	Italian (native) · English (C1)